What can I learn about compilers in one day?

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EJCP 2026 - https://ejcp2026-scilog.ls2n.fr/

École des jeunes chercheuses et jeunes chercheurs en Sciences du Logiciel

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Quick About Me: Guille

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- Pronounced *gi∫e* (guichet in FR, ~ghisheh in EN?)
- **Now:** Researcher at Inria - Lille
- Pharo Open Contributor since ~2010



- **Keywords:** compilers, testing, **software quality and its engineering**
- **Interests:** tooling, benchmarking, 日本語, board games, batman, concurrency, chess

Some months ago…



Guille, can you give a lecture about compilers?



Yes

Some months ago…



Guille, can you give a lecture about compilers **IN ONE DAY**?



I can try

Reality is Complex

- **Complex Domain + Short Time:**
  - Diving into a **real-world compiler** in a single day is not feasible
  - Lots of concepts, very technical

- **Unknown public:** you may have, maybe, perhaps, **experience. Or NOT.**

- We all **speak different languages**: french, spanish, *Java, Lisp, Smalltalk*…

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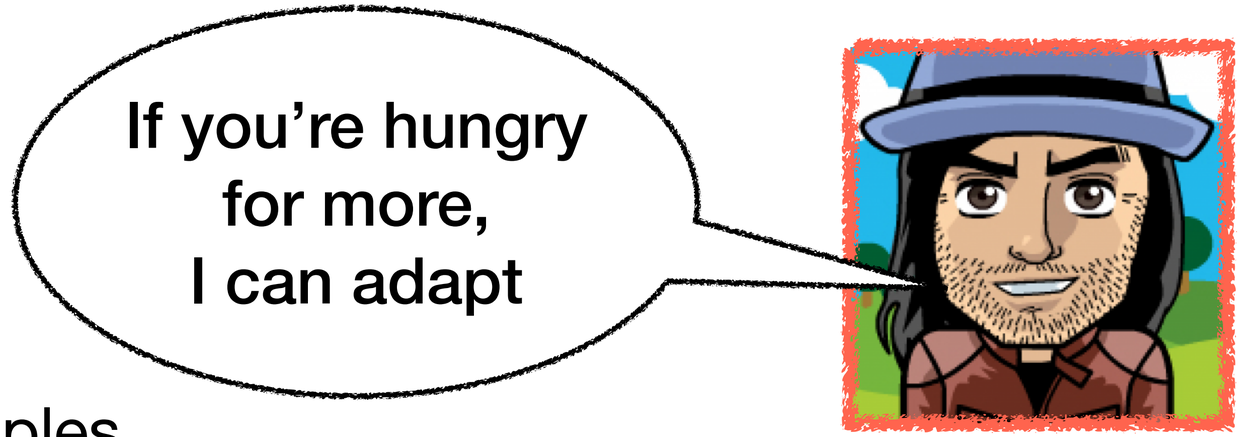
My Choice

- **Topics** we will discuss:
  - **Myths** like interpreted vs compiled languages
  - **Different machines**, and **compilation/translation** to those machines
  - **Testing** those translations

- **Exploration**-based labs:
  - Learn by observing real world examples

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Agenda Overview — We may deviate

- 9h-10h30 : **Theory —** Introduction to compilation and language implementations.
- 11h-12h30 : **Lab —** exploring generated code in different languages
- 14h-15h30 : **Theory —** Representing code, Compiler testing and Fuzzing
- 16h-17h30 : **Lab** — Fuzzing a buggy parser